

DMC

Definitions of
Channel
Capacity

Channel
Capacity of
Simple Channels

Iterative
Algorithms of
Computing
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ECE253/CSE208 Introduction to Information Theory

Lecture 10: Channel Capacity

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- Chap 7 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas

Discrete Memoryless Channels (DMC)

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We are seeking the answers to the following two core questions in information theory:

- The fundamental limit of **data compression** is $H(X)$ (Chapter 5).
- The fundamental limit of **data transmission** is channel capacity C (Chapter 7).

There is a duality between data compression and transmission.

- We want to **reduce redundancy** as much as possible for data compression.
- In the meantime, we add **controlled redundancy** to combat errors in data transmission.

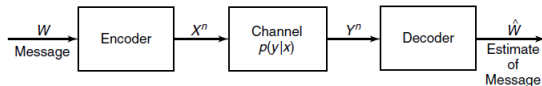


Figure: A generic diagram of communication systems.

Definitions of Discrete Memoryless Channel (DMC)

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Definitions of Channel Capacity

Channel Capacity of Simple Channels

Iterative Algorithms of Computing Channel Capacity

Definition (DMC)

A **discrete channel** is specified by an input alphabet $\mathcal{X}$, an output alphabet $\mathcal{Y}$, and a collection of transition probabilities

$$p(y | x), \quad x \in \mathcal{X}, y \in \mathcal{Y},$$

where $p(y | x)$ denotes the probability of observing output symbol y given that input symbol x is transmitted.

The channel is said to be **memoryless** if, for any sequence of inputs $x_1^n = (x_1, \dots, x_n)$ and outputs $y_1^n = (y_1, \dots, y_n)$,

$$p(y_1^n | x_1^n) = \prod_{i=1}^n p(y_i | x_i).$$

Equivalently, at each time i , the output Y_i depends only on the current input X_i and is conditionally independent of all past inputs and outputs.

Definitions of Channel Capacity

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Definition

The information-theoretical channel capacity of a DMC is defined as $C = \max_{p(x)} I(X; Y)$.

Definition

The operational channel capacity is the number of bits we can transmit reliably (with an arbitrarily low probability of error) per channel use.

$$C = \log(\text{number of reliably distinguishable inputs})$$

Shannon's second theorem (Noisy-channel coding theorem):

“Information-theoretical” channel capacity = “operational” channel capacity

Noiseless Binary Channel (NBC)

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From an operational perspective, we can transmit one bit without any error per channel use. Hence $C = 1$ bit. Meanwhile, we can calculate the *information-theoretical* capacity as

0 $\longrightarrow$ 0

X

Y

1 $\longrightarrow$ 1

$$C = \max_{p(x)} I(X; Y)$$

$$= \max_{p(x)} [H(X) - H(X|Y)]$$

$$= \max_{p(x)} H(X)$$

$$= 1 \text{ bit}, \quad \text{if } p(x) = (0.5, 0.5)$$

Noisy Channel with Non-overlapping Outputs

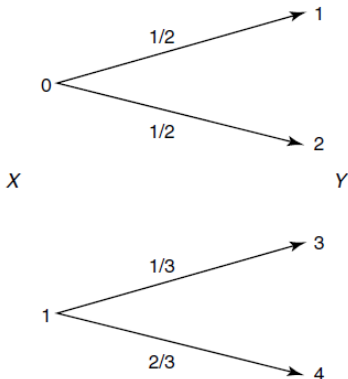
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Similar to the previous example, we can transmit one bit per channel use without error.
Hence $C = 1$ bit.



$$C = \max_{p(x)} I(X; Y)$$

$$= \max_{p(x)} [H(X) - H(X|Y)]$$

$$= \max_{p(x)} H(X)$$

$$= 1 \text{ bit,} \quad \text{if } p(x) = (0.5, 0.5)$$

[Exercise: solve it via $C = \max_{p(x)} (H(Y) - H(Y|X))$]

Noisy Typewriter

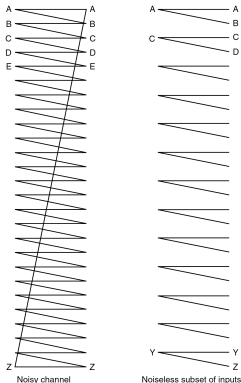
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The channel input is received correctly with probability 0.5, or is transformed into the next letter with probability 0.5. Note that if we transmit every other symbol $\{A, C, E, \dots, Z\}$, there will be no errors. Thus, $C = \log(13)$ bits.



$$\begin{aligned} C &= \max_{p(x)} I(X; Y) \\ &= \max_{p(x)} [H(Y) - H(Y|X)] \\ &= \max_{p(x)} [H(Y) - \sum p(x) H(Y|X = x)] \\ &= \max_{p(x)} [H(Y) - 1] \\ &= \log(13) \text{ bits, achieved when the input } X \text{ is uniform.} \end{aligned}$$

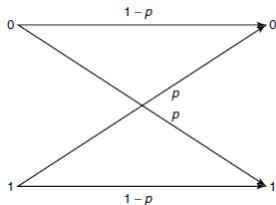
Binary Symmetric Channel (BSC)

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$$C = \max_{p(x)} I(X; Y)$$

$$= \max_{p(x)} [H(Y) - H(Y|X)]$$

$$= \max_{p(x)} [H(Y) - H(p)] \quad / H(Y|X = i) = H(p), i = 0, 1 /$$

$$= 1 - H(p) \text{ bits,} \quad \text{if } p(x) = (0.5, 0.5)$$

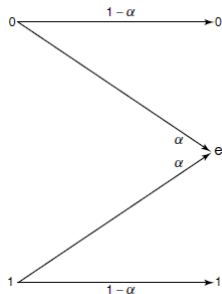
Binary Erasure Channel (BEC)

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$$C = \max_{p(x)} I(X; Y)$$

$$= \max_{p(x)} [H(Y) - H(Y|X)]$$

$$= \max_{p(x)} [H(Y) - H(\alpha)]$$

$$= \max_{\pi} (1 - \alpha) H(\pi)$$

$$= (1 - \alpha) \text{ bits, when } \pi := p(X = 1) = 0.5$$

Note that:

$$p(Y = 0) = (1 - \pi)(1 - \alpha), p(Y = 1) = \pi(1 - \alpha), p(Y = e) = (1 - \pi)\alpha + \pi\alpha = \alpha$$

$$\implies H(Y) = H((1 - \pi)(1 - \alpha), \pi(1 - \alpha), \alpha) = H(\alpha) + (1 - \alpha)H(\pi).$$

Symmetric Channel (SC)

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Consider the following doubly stochastic channel matrix

$$p(y|x) = \begin{pmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}.$$

All rows and columns are permutations of each other, the channel $p(y|x)$ is symmetric.

Capacity of Symmetric Channel

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$$\begin{aligned} C &= \max_{p(x)} H(Y) - H(Y|X) \\ &= \max_{p(x)} H(Y) - H(\mathbf{r}) \\ &= \log |\mathcal{Y}| - H(\mathbf{r}) \quad [\mathbf{r} \text{ denotes any row of the channel matrix}] \end{aligned}$$

$$\begin{aligned} p(y) &= \sum_x p(y|x)p(x) = \frac{1}{|\mathcal{X}|} \sum_x p(y|x) \\ &= \frac{c}{|\mathcal{X}|} = \frac{1}{|\mathcal{Y}|} \quad [\text{uniform input} \rightarrow \text{uniform output}] \end{aligned}$$

Weakly Symmetric Channel (WSC)

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Weakly symmetric channel:

If all rows are permutations of each other, while the sums of all columns $\sum_x p(y|x)$ are equal (not necessarily equal to 1); e.g.,

$$p(y|x) = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}.$$

Theorem

For a weakly symmetric channel, we have $C = \log |\mathcal{Y}| - H(\mathbf{r})$. This is achieved by a uniform distribution over $\mathcal{X}$.

Computing Capacity of General Channels

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Channel capacity has the following properties:

- Naive lower and upper bounds: $0 \leq C \leq \min(\log |\mathcal{X}|, \log |\mathcal{Y}|)$
- For a fixed channel $p(y|x)$, $I(X; Y)$ is a concave function of $p(x)$.

$$C = \max_{p(x) \in \mathcal{P}} I(X; Y),$$

where $\mathcal{P} = \{p(x) \mid 0 \leq p(x) \leq 1, \sum_x p(x) = 1\}$ denotes the probabilistic simplex.

Generally, no closed-form solution for the above nonlinear optimization problem. We can leverage various algorithms to numerically compute C ; e.g.,

1. Gradient search based algorithms
2. KKT conditions
3. Other iterative algorithms

Projected Gradient Ascent

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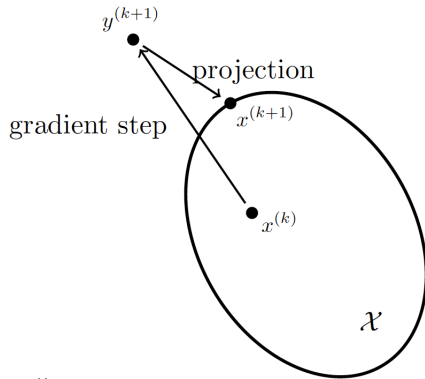
$$C = \max_{p(x) \in \mathcal{P}} I(X; Y)$$

$$= \max_{p(x) \in \mathcal{P}} \sum_{x, y} p(x) p(y|x) \log \frac{p(x) p(y|x)}{\sum_{x'} p(x') p(y|x')}$$

Projected gradient ascent:

$$p^{\ell+1}(x) = \text{Proj}_{\mathcal{P}} [p^{\ell}(x) + \alpha^{\ell} \nabla f(p^{\ell}(x))] ,$$

where ℓ is the iteration index; α^{ℓ} is the step size; and $\nabla f(\cdot)$ is the gradient of the objective function.



Blahut-Arimoto Algorithm (1/2)¹

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Blahut-Arimoto algorithm: Treat $p(x)$ and $p(x|y)$ as independent variables.

$$C = \max_{p(x)} I(X; Y) = \max_{p(x)} \sum_{x,y} p(x) p(y|x) \log \frac{p(x|y)}{p(x)} \quad (1)$$

$$= \max_{p(x)} \max_{q(x|y)} \sum_{x,y} p(x) p(y|x) \log \frac{q(x|y)}{p(x)} \quad (2)$$

¹S. Arimoto (1972), "An algorithm for computing the capacity of arbitrary discrete memoryless channels".

R. Blahut (1972), "Computation of channel capacity and rate-distortion functions".

Blahut-Arimoto Algorithm (2/2)

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For fixed $p(x)$, update $q(x|y)$ as

$$q(x|y) = \frac{p(x)p(y|x)}{\sum_{x'} p(x')p(y|x')}.$$

For fixed $q(x|y)$, update $p(x)$ as

$$p(x) = \frac{\prod_y q(x|y)^{p(y|x)}}{\sum_{x'} \prod_y q(x'|y)^{p(y|x')}}.$$

The above closed-form updates are justified by the two lemmas given below.

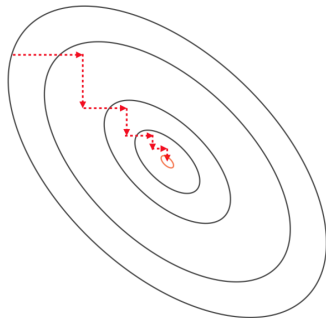


Figure: For block coordinate ascent, we only update one block at each iteration:
 $p^1(x) \rightarrow q^1(x|y) \rightarrow p^2(x) \rightarrow q^2(x|y) \rightarrow \dots$

Lemma 1²

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Here we show two lemmas to validate the Blahut-Arimoto algorithm. Let

$$J(p) \triangleq \sum_{x,y} p(x)p(y|x) \log \frac{p(x|y)}{p(x)} \quad (3)$$

$$J(p, q) \triangleq \sum_{x,y} p(x)p(y|x) \log \frac{q(x|y)}{p(x)} \quad (4)$$

Lemma

For any fixed p , $J(p, q) \leq J(p)$, with equality iff $q(x|y) = \frac{p(x)p(y|x)}{\sum_{x'} p(x')p(y|x')} = p(x|y)$.

Proof:

$$J(p) - J(p, q) = \sum_{x,y} p(x)p(y|x) \log \left[\frac{p(x|y)}{q(x|y)} \right] = D(p(x|y) || q(x|y)) \geq 0.$$

²http://ecse.rpi.edu/~pearlman/lec_notes/arimoto_2.pdf

Lemma 2

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Lemma

For any fixed $q(x|y)$, $J(p, q) \leq \log(\sum_x r(x))$, where $r(x) = \exp\left(\sum_y p(y|x) \log q(x|y)\right)$
with equality if and only if $p(x) = \frac{r(x)}{\sum_{x'} r(x')}$.

Proof Lemma 2

Let $r(x) = \exp \left(\sum_y p(y|x) \log q(x|y) \right)$ and $\tilde{r}(x) \triangleq \frac{r(x)}{\sum_x r(x)}$. Then, we have

$$\begin{aligned} J(\mathbf{p}, \mathbf{q}) &= \sum_{x,y} p(x)p(y|x) \log \frac{1}{p(x)} + \sum_{x,y} p(x)p(y|x) \log q(x|y) \\ &= \sum_x p(x) \log \left(\frac{\exp \sum_y p(y|x) \log q(x|y)}{p(x)} \right) \\ &= \sum_x p(x) \log \left(\frac{\tilde{r}(x)}{p(x)} \times \sum_x r(x) \right) \\ &= \sum_x p(x) \log \left(\sum_x r(x) \right) - D(p||\tilde{r}) \leq \log \left(\sum_x r(x) \right) \end{aligned}$$

with equality iff $p(x) = \tilde{r}(x) = \frac{\prod_y q(x|y)^{p(y|x)}}{\sum_{x'} \prod_y q(x'|y)^{p(y|x')}}.$